

1- Introduction to Bayesian Methods

Anabel Blasco

Department de Matemàtiques

Universitat Autònoma de Barcelona

Anabel.blasco@uab.cat

Overview

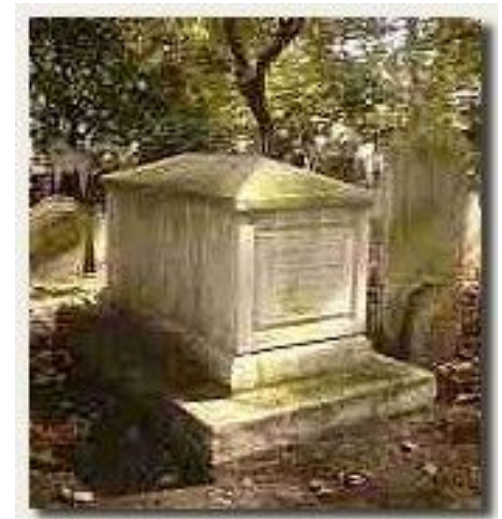
- 1- Bayes' theorem and its consequences.
- 2- The basics of Bayesian Statistics: prior distributions.
- 3- Bayesian inference: the posterior distribution.



[Thomas Bayes](#) 1702-1761

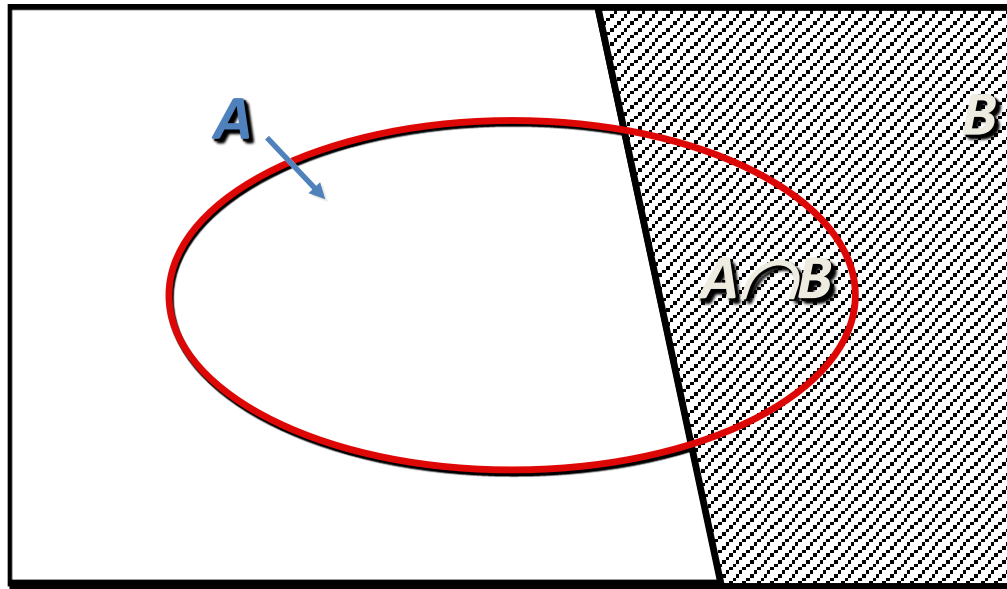
Two years after his death, "An Essay Towards Solving a Problem in the Doctrine of Chances" by English clergyman and mathematician Thomas Bayes, was published in the Philosophical Transactions of the Royal Society 53 (1763) 370-418. This paper was sent to the Royal Society by Bayes's friend Richard Price. [Bayes's paper enunciated Bayes's Theorem for calculating "inverse probabilities".](#)

In recognition of Thomas Bayes's important work in probability. The vault was restored in 1969 with contributions received from statisticians throughout the world.



Conditioned probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



$$P(A \cap B) = P(B)P(A|B)$$

Conditioned probability incorporates the *prior* information (B) to calculate the probability of A.

Enjoy with these exercises!



- Mr. Jones has two children. What is the probability that both children are girls? (no prior information)
- Mr. Jones has two children. At least one of them is a girl. What is the probability that both children are girls?
- Mr. Jones has two children. The older child is a girl. What is the probability that both children are girls?
- Read the history of the professor of statistics carrying a bomb in his baggage:

<http://www.onlinemathlearning.com/math-jokes-statistics.html>

Enjoy with these exercises!



- We are to play a version of Russian Roulette, the revolver is a standard six shooter but I will put two bullets in the gun in consecutive chambers. I spin the chambers, put the gun to my head pull the trigger and survive. I hand you the gun and give you a choice... You may put the gun straight to your head and pull the trigger, or you may re-spin the gun before you do the same.

What is the better choice in order to survive?

1- Bayes' theorem

Using the expressions of conditioned probability,

$$P(A \cap B) = P(A)P(B | A)$$

$$P(A \cap B) = P(B)P(A | B)$$

and combined them,

$$P(A)P(B | A) = P(B)P(A | B)$$

$$P(A | B) = \frac{P(A)P(B | A)}{P(B)}$$

Frequently, the probability of the denominator is calculated using the law of total probability, obtaining this expression:

$$P(A | B) = \frac{P(A)P(B | A)}{P(A)P(B | A) + P(A^c)P(B | A^c)}$$

Exercise: The result of throwing a fair coin can be head or tail, with probability $p_{\text{head}}=1/2$. If the coin is loaded $p_{\text{head}}=3/4$. Suppose 99% of the coins are fair. What is the probability that a given coin is fair if launching it 3 times one gets 3 heads?

Example

- A **diagnostic test** helps to calculate the probability for an individual to have a disease.
- We know initially the value of $P(\text{sick})$ because this is the prevalence of the disease.
 - **Prevalence**: The proportion of individuals in a population having a disease.
 - We use a diagnostic or clinical test to aid in the diagnosis or detection of disease. The result of the diagnostic test can be “positive” (+) or “negative” (-). Each test is evaluated, and the following parameters of quality are known:
 - $P(+ \mid \text{sick}) = \text{Sensitivity}$ = the ability of the test to identify positive results
 - $P(- \mid \text{healthy}) = \text{Specificity}$ = the ability of the test to identify negative results
 - Using Bayes’ theorem we can calculate the predictive values of the diagnostic test: $P(\text{sick} \mid +)$ and $P(\text{healthy} \mid -)$.

The presence of glucosuria in an urine analysis is an indicator of diabetes.

- Diabetes affects to 2% of the population.
- The sensibility of the glucosuria-diagnostic test is 0.945.
- The specificity is 0.977.

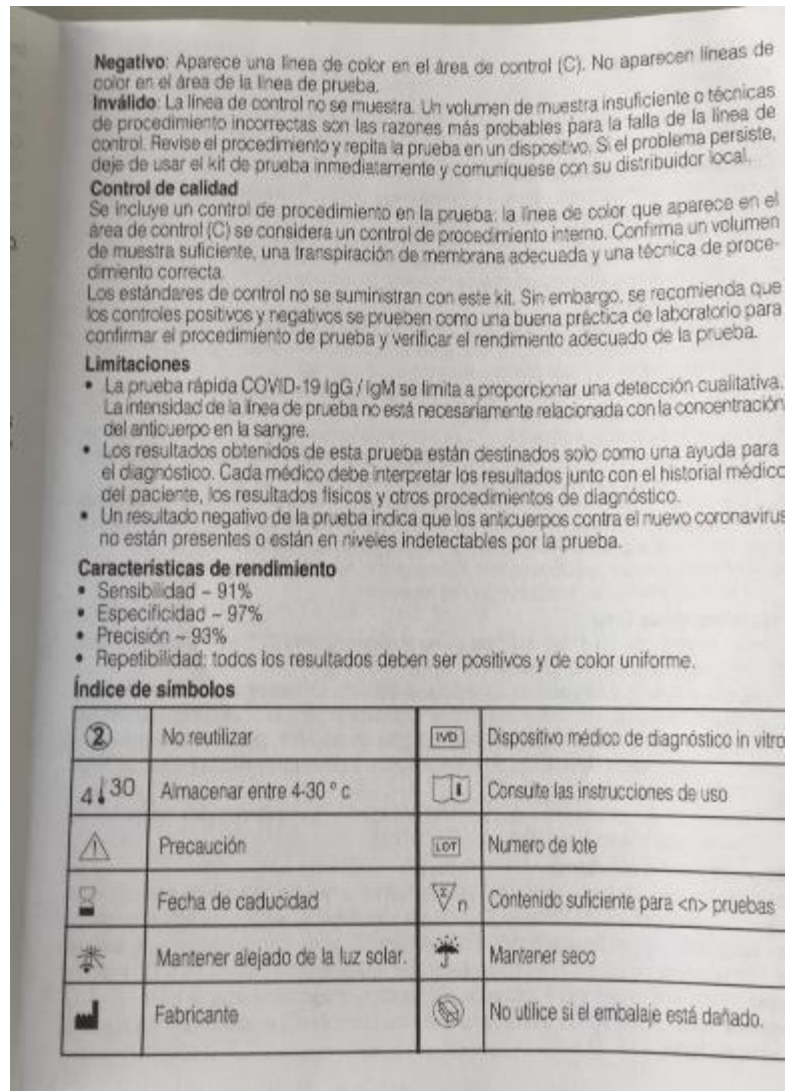


Exercise: Using Bayes' theorem show that the predictive values are $P(\text{sick} \mid +) = 0.456$ and $P(\text{healthy} \mid -) = 0.999$

Is this a good diagnostic test?



Rapid test for SARS-CoV-2 diagnosis



Exercise: Compute $P(\text{sick} \mid +)$ and $P(\text{healthy} \mid -)$ using the values of the estimated prevalence of SARS-CoV-2 in Barcelona (7% approx.)

[https://www.mscbs.gob.es/ciudadanos/en-covid/docs/ESTUDIO ENE-COVID19 INFORME FINAL.pdf](https://www.mscbs.gob.es/ciudadanos/en-covid/docs/ESTUDIO_ENE-COVID19_INFORME_FINAL.pdf)

The Bayesian thinking

- Before (prior) the glucosuria-diagnostic test, the chance of a patient to have diabetes is just 2% (the population prevalence).
- After (posterior) the new data (the result of the diagnostic test has been +), the chance is 0.456

Bayesian methods learn from data. The role of the experiment is to add to our knowledge and so to update what we can say about the parameters.

Bayesian statistics versus non-bayesian

Statistics estimates unknown parameters (like the mean or the variance of a population). Parameters represent things that are unknown. They are some properties of the population from which the data arise. Questions of interest are expressed as questions on such parameters: confidence intervals, hypothesis testing etc.

Classical (frequentist) statistics considers parameters as specific to the problem, so that they are not subject to random variability. Hence parameters are just unknown constants, they are not random, and it is not possible to make probabilistic statements about parameters. (Remember the meaning of a CI)

Conversely, Bayesian statistics considers parameters as random and hence it is allowed to make probabilistic statements about them. The distribution of the parameters is called *prior distribution*, expressing what is known (or believed to be true) before seeing the data.

2-The prior distribution

Approaches to choose a prior distribution:

- **Informative prior distribution.** We have to use the knowledge about the problem, perhaps based on other data. It is necessary to construct a distribution reflecting all beliefs about the unknown parameters.
- **Noninformative prior distribution.** This is implemented by assigning equal probability to all possible values of the parameters (uniform distributions).

The prior distribution: an example

(from *Basic Bayesian Methods*, Mark E. Glickman and David A. van Dyk)

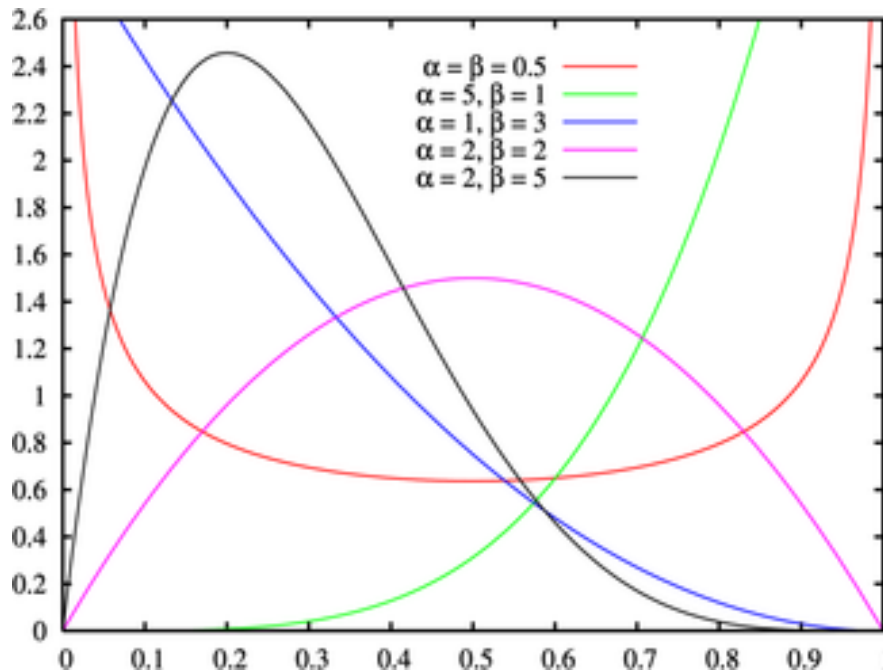


We are interested in the probability θ , that a randomly selected woman, aged 60–69 years with a family history of cancer, had a positive breast cancer screening. Because the parameter θ belongs to $[0,1]$, we have to define a prior distribution over this interval. What we know *a priori* about θ ?

If we do not trust in the information provided by the American Cancer Society, a natural noninformative prior distribution is the uniform over $[0,1]$. But an expert tell us, “Anyway I’m sure that the value of θ can not be greater than 5%”. Then we could define a kind of noninformative prior distribution using the uniform over $[0,0.05]$.

However, according to the American Cancer Society, roughly 3.6% of women aged 60–69 years develop invasive breast cancer, so that we can construct an informative prior distribution for θ reflecting this information. In this case, a flexible choice could be the **Beta distribution**.

Beta densities (https://en.wikipedia.org/wiki/Beta_distribution)



$$\begin{aligned}
 f(x; \alpha, \beta) &= \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\int_0^1 u^{\alpha-1}(1-u)^{\beta-1} du} \\
 &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1} \\
 &= \frac{1}{B(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1}
 \end{aligned}$$

The expectation and variance are,

$$E(X) = \frac{\alpha}{\alpha + \beta}$$

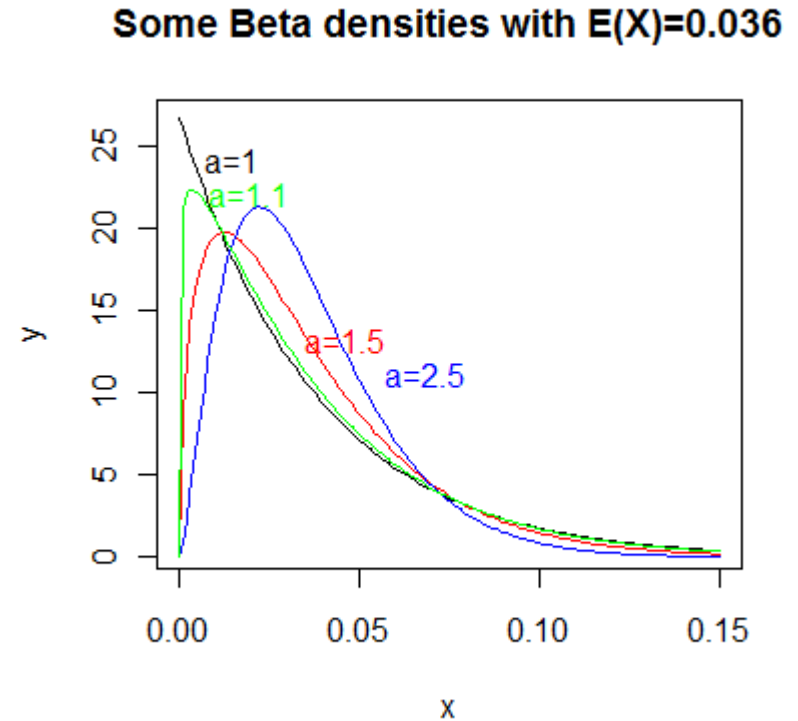
$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

We can take $\beta = 26.7778\alpha$ and then $E(X) = 0.036$. Using values $\alpha > 1$ the density has a mode.


```

x=seq(0,0.15,0.001)
y=dbeta(x,1,1*26.7778)
y2=dbeta(x,1.5,1.5*26.7778)
y3=dbeta(x,2.5,2.5*26.7778)
y4=dbeta(x,1.1,1.1*26.7778)
plot(x,y,type="l")
lines(x,y2,col="red")
lines(x,y3,col="blue")
lines(x,y4,col="green")
legend(0.002,26,"a=1",bty="n")
legend(0.052,13,"a=2.5",text.col="blue",bty="n")
legend(0.03,15,"a=1.5",text.col="red",bty="n")
legend(0.003,24,"a=1.1",bty="n",text.col="green")
title(main = Some Beta densities with E(X)=0.036)

```



3- Bayesian Inference: the posterior distribution

Once the data has been observed, the likelihood function is constructed. This is the joint probability function of the data, but viewed as a function of the parameters. Assuming that the data values, $X = (x_1, \dots, x_n)$ are obtained independently, the likelihood function is given by,

$$L(X|\theta) = p(x_1, \dots, x_n|\theta) = \prod_{i=1}^n p(x_i|\theta)$$

where $p(x|\theta)$ is the density or probability function for each observation.

To obtain the *posterior distribution (or density)*, $f(\theta | X)$, the probability distribution of the parameters once the data have been observed, we apply Bayes' theorem.

The posterior distribution

$$f(\theta | X) = \frac{\pi(\theta)L(X | \theta)}{\int_{-\infty}^{\infty} \pi(\theta)L(X | \theta)d\theta}$$

Here $\pi(\theta)$ is the prior density. Note that the denominator is just a normalizing constant not depending on the parameter.

It is straightforward to obtain the posterior distribution: Simply multiply the prior distribution by the likelihood, and then determine the constant that forces the expression to integrate to 1.

From the posterior density we can estimate θ by means of the expectation (or median, or mode), and we can calculate confidence intervals, called *credibility intervals*, using the percentiles.

Exercise: Suppose that the posterior density is unimodal.

- How to find a credibility interval of size 95% with minimum length?
- The answer is also useful for computing confidence intervals with minimum length in frequentist inference.



An example (continued)

(from *Basic Bayesian Methods*, Mark E. Glickman and David A. van Dyk)



Suppose, for the breast cancer screening study, 14 of the 300 women had positive tests. Thus 14 women have $x_i = 1$, and the remaining 286 have $x_i = 0$. The likelihood is therefore given by, $L(X|\theta) = \theta^{14}(1-\theta)^{286}$

Taking the Beta prior distribution with parameters $\alpha=2.5$ and $\beta=66.9445$, the following posterior density is obtained: $f(\theta|X) = c \theta^{15.5}(1-\theta)^{351.9445}$

Thus, the posterior distribution is Beta(16.5, 352.9445).

Then we can obtain point estimates of the parameter in the following way:

1- Expectation	$\theta_1^* = 0.0447$	$\alpha/(\alpha+\beta)$
2- Mode	$\theta_2^* = 0.0422$	$(\alpha-1)/(\alpha+\beta-2)$
3- Median	$\theta_3^* = 0.0438$	<code>qbeta(0.5,16.5,352.9445)</code>
4- Classical (non-Bayesian)	$\theta_4^* = 0.0467$	

An example (continued)



The likelihood is therefore given by, $L(X|\theta) = \theta^{14}(1-\theta)^{286}$

Taking the uniform[0,1] prior noninformative distribution, the following posterior density is obtained: $f(\theta|X) = c \theta^{14}(1-\theta)^{286}$

Thus, the posterior distribution is Beta(15 , 287).

Then we can obtain point estimates of the parameter in the following way:

1- Expectation	$\theta^*_1 = 0.0497$	$\alpha/(\alpha+\beta)$
2- Mode	$\theta^*_2 = 0.0467$	$(\alpha-1)/(\alpha+\beta-2)$
3- Median	$\theta^*_3 = 0.0487$	$qbeta(0.5,15,287)$
4- Classical (non-Bayesian)	$\theta^*_4 = 0.0467$	

“Confidence intervals” can be calculated using the percentiles. For example, a 95% central posterior interval involves computing the 2.5% and 97.5% percentiles of the posterior distribution. Taking the uniform[0,1] noninformative distribution, the corresponding percentiles are,

```
qbeta(c(0.025,0.975),15,287)  
[1] 0.02815644 0.07680793
```

Thus, a 95% credibility interval for θ is [0.0281 , 0.0768] .

Taking the Beta prior distribution with parameters $\alpha=2.5$ and $\beta=66.9445$, the percentiles are now,

```
qbeta(c(0.025,0.975),16.5,352.9445)  
[1] 0.02606071 0.06791458
```

Now, a 95% credibility interval for θ is [0.0261 , 0.0679]

The classical 95% confidence interval is,

$$0.0467 \pm 0.0238 = [0.0229, 0.0705]$$

Interpretation: There is a 95% chance that the proportion θ lies between 2.3% and 7.1%.

Correct? NO!



However, in the Bayesian context, parameters are random and when we compute a Bayesian interval for θ it means exactly the interpretation usually wrongly given to a confidence interval.

Summary: the steps to perform Bayesian inference.

1. Formulate a probability model for the data.
2. Decide on a prior distribution, which quantifies the uncertainty in the values of the unknown model parameters before the data are observed.
3. Observe the data, and construct the likelihood function based on the data and the probability model formulated in step 1. The likelihood is then combined with the prior distribution from step 2 to determine the posterior distribution, which quantifies the uncertainty in the values of the unknown model parameters after the data are observed.
4. Summarize important features of the posterior distribution, or calculate quantities of interest based on the posterior distribution. These quantities constitute statistical outputs, such as point estimates and intervals.

Exercise: Make Bayesian inference for the breast cancer screening study.

- Use the opinion of the expert, “Anyway I’m sure that the value of θ can not be greater than 5%”, and use as a prior distribution the uniform over $[0, 0.05]$.
- Calculate the posterior density and find point estimates of parameter θ .
- Calculate a 95% centered credibility interval for θ .

